

Introduction to Machine Learning

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Outline

- ① Introduction to Machine Learning
- ② Relationship Between Machine Learning, Statistics, and Data Science
- ③ Types of Machine Learning
- ④ Applications of Machine Learning

Introduction to Machine Learning

Machine Learning Is Already Part of Your Day

- Your phone recognizes your face to unlock itself.
- Netflix and YouTube suggest what to watch next based on your viewing history.
- Gmail quietly moves suspicious messages into your spam folder.
- Google Maps predicts how long your commute will take, using real-time and historical traffic data.
- Your bank flags an unusual transaction on your card as possible fraud.

All of these are machine learning systems working in the background — none of them were explicitly programmed with a rule for every situation; they learned from data.

What is Machine Learning?

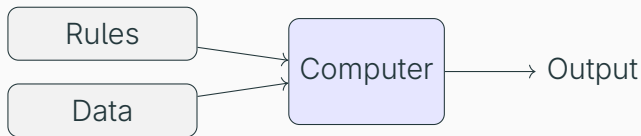
Tom Mitchell's formal definition: a program learns from experience E with respect to task T and performance measure P , if its performance at T , measured by P , improves with E .

Example (spam filter):

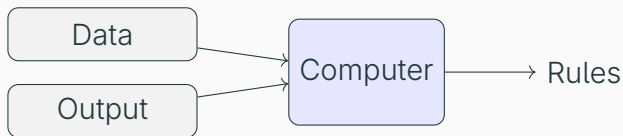
- E (experience): a collection of past emails, each already labeled spam or not spam.
- T (task): classifying a new, incoming email as spam or not spam.
- P (performance measure): the percentage of emails correctly classified.
- As the filter sees more labeled emails (E), its classification accuracy (P) on new emails improves — that improvement *is* the learning.

Traditional Programming vs. Machine Learning

Traditional Programming



Machine Learning



Instead of a human writing explicit rules, we give the algorithm examples (data), and it discovers the rules (a **model**) on its own.

Inputs, Outputs, and Terminology

- **Motivating Example:** Predict product sales (Y) based on advertising budgets spent across three media: *TV* (X_1), *radio* (X_2), and *newspaper* (X_3).
- **Input Variables (X):**
 - Represented as a vector $X = (X_1, X_2, \dots, X_p)$.
 - Synonyms: **predictors**, **independent variables**, **features**, or **covariates**.
- **Output Variable (Y):**
 - The target quantity we wish to estimate or predict.
 - Synonyms: **response** or **dependent variable**.

Formulating the Problem: $Y = f(X) + \epsilon$

We assume a general relationship between the response Y and predictors X :

$$Y = f(X) + \epsilon$$

- **Systematic Function (f):**

- A fixed but *unknown* function connecting $X_1, \dots, X_p$ to Y .
- Represents the systematic information that inputs provide about the output.

- **Random Error Term (ϵ):**

- Captures unmeasured factors, noise, or inherent variation.
- Assumed to be independent of X with a mean of zero ($E[\epsilon] = 0$).

The Core Goal: Estimating f

- Because the true function f is unknown, we must use observed data to compute an estimate $\hat{f}$.
- **Statistical / Machine Learning:** A collection of data-driven approaches for estimating f to obtain predictions:

$$\hat{Y} = \hat{f}(X)$$

- **Two Main Goals of Estimating f :**
 - ① **Prediction:** Make accurate predictions of $\hat{Y}$ for new, unseen observations of X .
 - ② **Inference:** Understand the relationship between X and Y (e.g., *Which media budgets actually drive sales?*).

Relationship Between Machine Learning, Statistics, and Data Science

Statistics

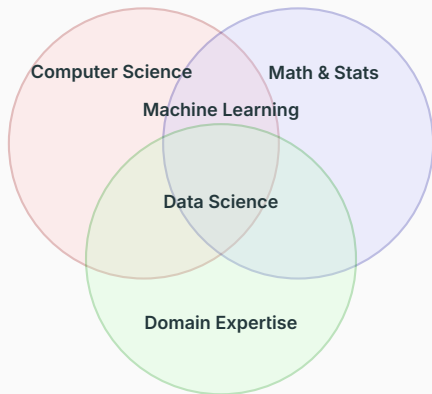
- **Statistics:** the mathematical discipline of collecting, analyzing, and interpreting data, with heavy emphasis on *inference* — quantifying uncertainty, testing hypotheses, estimating confidence intervals.
- Emphasis on mathematical inference, hypothesis testing, and quantifying uncertainty.
- A statistician cares whether a coefficient is significant and interpretable; an ML engineer mostly cares whether the model predicts well on new data, even if it is a black box.

Machine Learning

- **Machine Learning:** algorithms that learn patterns from data primarily to make *predictions* on new, unseen data.
- Emphasis on predictive algorithms that generalise well to unseen data.
- Statistics supplies much of ML's theoretical foundation (linear regression, maximum likelihood estimation, hypothesis testing).
- ML is one tool a data scientist uses, alongside dashboards, ETL pipelines, and A/B testing.

Data Science

- **Data Science:** a broad, applied field that combines domain knowledge, programming/engineering, statistics, and ML to go from raw data to business decisions — includes data collection, cleaning, visualization, and communicating results to stakeholders.
- An interdisciplinary umbrella combining domain knowledge, software engineering, stats, and ML to solve end-to-end problems.
- A data scientist at a company typically needs a bit of both statistics and ML, as well as the ability to explain results to a non-technical manager.



Types of Machine Learning

Supervised Learning

Learning a mapping from input **features** X (the measurable inputs, e.g., square footage, pixel values) to an output **label** Y (the “correct answer”), using a **training set** of examples where Y is already known.

- **Classification:** label is a discrete category. Example: is this email spam or not? Does this X-ray show a tumor or not (used in hospital diagnostic - support tools)?
- **Regression:** label is a continuous number. Example: predicting a house's sale price from its size, location, and age; predicting tomorrow's temperature.
- The **model** is the learned function $\hat{Y} = f(X)$; a **loss function** measures how wrong its predictions are, and training adjusts the model to minimize this loss on the training data while still **generalizing** well to new, unseen data (avoiding **overfitting**, i.e., memorizing training data instead of learning general patterns).

Supervised Learning: Examples

- Everyday examples: your email's spam filter, your phone unlocking with face recognition, autocorrect/predictive text.
- Industry examples: banks predicting loan default risk (credit scoring), Tesla/Waymo detecting pedestrians and stop signs from camera images, hospitals using ML to flag likely - cancerous scans for radiologist review.

Unsupervised Learning

Finding structure or patterns in data that has **no labels** — there is no “correct answer” provided, the algorithm must discover structure on its own.

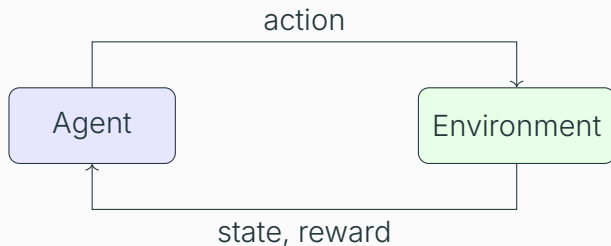
- **Clustering**: grouping similar data points together (e.g., k -means). Example: a retailer segments customers into groups by purchasing behavior, without being told in advance what the groups should be, in order to target each group with different marketing.
- **Dimensionality reduction**: compressing many features into a few that capture most of the information (e.g., PCA), useful for visualization or as a preprocessing step.
- **Anomaly detection**: flagging data points that don't fit the usual pattern. Example: banks flagging an unusual credit card transaction as possible fraud.

Unsupervised Learning: Examples

- Google Photos automatically grouping photos of the same person's face together, even though it was never told who that person is.
- Genomics researchers clustering gene-expression data to discover previously unknown subtypes of a disease.

Reinforcement Learning

An **agent** learns to make a sequence of decisions by interacting with an **environment**: at each **state**, it takes an **action**, receives a **reward** (or penalty), and updates its **policy** (its strategy for choosing actions) to maximize cumulative reward over time.



Reinforcement Learning (cont.)

- Unlike supervised learning, there is no fixed “correct” action for each state — the agent must learn from trial and error, and rewards can be delayed (an action now may only pay off many steps later).
- Analogy: training a dog with treats — you don’t tell it the exact muscle movements for “sit,” you reward it when it happens to get closer to the desired behavior.
- Example: DeepMind’s AlphaGo learned to play Go well enough to beat the world champion, largely through self-play reinforcement learning.

Reinforcement Learning: Examples

- Robots learning to walk or grasp objects through repeated physical trial and error.
- Google used reinforcement learning to reduce the energy used for cooling its data centers.
- YouTube uses RL-based methods to optimize recommendations for long-term viewer engagement rather than just the next click.

Semi-Supervised Learning

Semi-Supervised Learning: Bridges the gap between supervised and unsupervised learning by leveraging a *small set* of labeled data alongside a *large pool* of unlabeled data.

- Drastically reduces human labeling cost while boosting model accuracy.
- *Example:* Medical diagnostics where only a tiny fraction of MRI scans are labeled by expert radiologists.

Self-Supervised Learning

Self-Supervised Learning: The model generates its own labels directly from the input data structure without human annotation.

- Hides part of the data and trains the model to predict the missing pieces (pretext task).
- Powers modern Large Language Models (LLMs) and foundation vision models.
- *Example:* Training ChatGPT by predicting the next word in billions of sentences, or predicting masked patches in images.

Applications of Machine Learning

Applications in Everyday Life

- **Voice assistants:** Siri, Alexa, and Google Assistant use ML for speech recognition and natural language understanding.
- **Recommendations:** Netflix suggesting what to watch, Spotify's Discover Weekly, YouTube's homepage, Amazon's "customers also bought."
- **Navigation:** Google Maps predicting traffic and estimated arrival time from historical and real-time location data.
- **Photography and communication:** face unlock on smartphones, autocorrect and predictive text, spam filtering in email, automatic photo tagging.
- **Social media:** automatically filtering out content, ranking your feed, and suggesting people you may know.

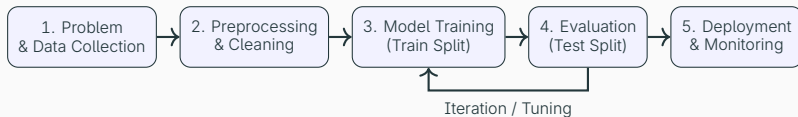
Applications in Industry and Organizations

- **Healthcare:** diagnostic support from medical images (X-rays, MRIs); DeepMind's AlphaFold predicting 3D protein structures, accelerating drug discovery.
- **Finance:** fraud detection on card transactions, algorithmic trading, automated credit scoring for loan approval.
- **Manufacturing:** predictive maintenance — companies like Siemens and GE use sensor data to predict when a machine will fail *before* it breaks down, avoiding costly downtime.
- **Retail:** demand forecasting and inventory management (e.g., Amazon predicting how much of a product to stock in each warehouse).

Applications in Industry and Organizations (cont.)

- **Transportation:** self-driving vehicles (Tesla, Waymo), and dynamic/surge pricing for ride-sharing (Uber, Lyft) based on real-time demand.
- **Agriculture:** predicting crop yields and detecting plant disease from drone/satellite imagery for precision farming.
- **Environmental Science:** weather forecasting and climate change modeling.

The Machine Learning Lifecycle



- **Train/Test Split:** Data is partitioned into a *training set* (used to fit the model) and a *test set* (held out to evaluate true generalization).
- **Iterative Cycle:** Real-world ML involves continuous retraining as model performance degrades on new incoming data over time.

Questions?
